APPLIED MATHEMATICS & TECHNICAL FORMULAE HANDBOOK

Essential Engineering Calculations: Ohm Law, Power, Cable Sizing, BTU Cooling Capacity

1. FUNDAMENTAL ELECTRICAL FORMULAE

OHM LAW & DC / SINGLE PHASE AC POWER:

* Voltage: V = I x R | Current: I = V / R | Resistance: R = V / I
* Active Power (Watts): P = V x I x Cos(phi) = I^2 x R = V^2 / R
* Apparent Power (Volt-Amperes): S = V x I [VA or kVA]
* Reactive Power (VAR): Q = V x I x Sin(phi) [VAR or kVAR]
* Power Factor: PF = Cos(phi) = Real Power (kW) / Apparent Power (kVA)

2. THREE-PHASE MOTOR CALCULATIONS

STAR AND DELTA BALANCED FORMULAE:

* Star Connection (Y): Line Voltage VL = sqrt(3) x Vph | Line Current IL = Phase Current Iph
* Delta Connection (D): Line Voltage VL = Phase Voltage Vph | Line Current IL = sqrt(3) x Iph
* Three Phase Active Power: P = sqrt(3) x VL x IL x Cos(phi)
* Synchronous Speed: Ns = (120 x Frequency) / Number of Poles = (120 x 50) / 4 = 1500 RPM
* Motor Slip (%): Slip = [(Ns - Actual Speed) / Ns] x 100

3. HVAC & REFRIGERATION COOLING LOAD CALCULATIONS

TON OF REFRIGERATION (TR) & HEAT CAPACITY:

* 1 Ton of Refrigeration (TR) = 12,000 BTU/hr = 3,024 kcal/hr = 3.517 kW (Thermal)
* Sensible Heat: Q_sensible = m x c x Delta_T [Joules or Watts]
* Latent Heat: Q_latent = m x L (L_vaporization for water = 2260 kJ/kg)
* Quick Room Sizing: Room Area (sq ft) x 50 to 60 BTU/sq ft (e.g. 150 sq ft x 60 = 9,000 BTU -> 1.0 Ton)